IDENTIFICATION OF A TARGETED SEARCH MODEL FOR TWO-SIDED MATCHING MARKETS

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ABSTRACT. Matching among similar people may reflect preferences or the composition of their social circles. I establish conditions for identifying preference and social-circle parameters from observed search behavior. Building on Cheremukhin et al. (2020, 2015), I develop a two-sided search model with unobserved payoff heterogeneity, allowing the value of a match to vary across individuals. Individuals choose how likely they are to contact each partner type, such as an education group. Social circles determine the probabilities of contacting each type without additional effort. A targeting cost captures the effort required to redirect search. Matches require reciprocal search. I establish equilibrium existence and show that social circles can generate matching among similar partners when targeting costs are high, even without a preference for similarity. Individual contact probabilities identify targeting costs through quantile comparisons that use a specified distribution of unobserved payoff heterogeneity. Observed social circles then allow recovery of preferences. When circles are unobserved, I give conditions under which changes in targeting costs identify both preferences and social-circle composition. When only average contact probabilities are observed for each type, I compare environments with different social circles but common preferences and targeting costs, and establish conditions for local identification of preferences and targeting costs.

KEYWORDS. Two-sided matching; targeted search; social circles; identification.

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1 Introduction

Finding a partner often begins with the people one already knows. Friends, colleagues, and family connections shape whom a person is likely to meet, and these contacts often bring together people with similar backgrounds. Someone whose social circle consists largely of university graduates, for example, may be more likely to meet a prospective partner with a university degree. Reaching beyond familiar circles can require additional effort, such as attending new social events, seeking introductions, or travelling to meet people elsewhere. Who ultimately matches with whom may therefore reflect preferences for particular partners, the opportunities provided by existing social connections, and the cost of searching beyond them.

This raises a central question: to what extent do observed matching patterns reflect preferences, and to what extent do they reflect social circles? An analyst seeking to recover preferences and search costs must account for the opportunities provided by social circles. I develop a targeted search model to establish conditions for recovering preferences, targeting costs, and, when unobserved, social-circle composition from search behavior. The identifying data consist of individual contact probabilities or their averages within observed groups, together with information about social circles or changes in search costs.

The model builds on the targeted search framework of Cheremukhin et al. (2020) and the probabilistic-choice framework of Cheremukhin et al. (2015). Partner types are defined by observed characteristics such as education. An individual's social circle determines a *reference distribution*: the probabilities of contacting each type without additional effort. If 70 percent of routine contacts are university graduates, for example, a search contact reaches a graduate with probability 0.7 at the baseline. The individual can raise this probability by seeking introductions to graduates or attending alumni events. The resulting probabilities of contacting different types are the *targeting probabilities*. Departures from the baseline incur a *targeting cost*, representing the time and resources required to change whom one is likely to meet. I use relative entropy to measure how much the chosen probabilities depart from the baseline. Individuals choose their contact probabilities by comparing these costs with the gains from matching and the chance that a prospective partner reciprocates.

I incorporate unobserved payoff heterogeneity into this framework, allowing individuals with the same observed characteristics to value potential matches differently. Relative-entropy costs yield explicit best responses, and I establish an equilibrium in which both sides' search decisions are mutually optimal. Match payoffs are taken as given. As targeting costs grow, contact probabilities approach social-circle composition. When similar types are common in each other's circles, reciprocal search can therefore generate positive assortative matching—matching between similar types—even without a preference for similarity. A two-type example makes this mechanism explicit.

The identification analysis considers three data settings. Each uses a specified distribution of unobserved payoff heterogeneity, including its location and scale.

First, suppose the researcher observes each individual's probabilities of contacting different partner types and the social-circle composition shared by individuals of the same observed type. Holding the probabilities of successful contact fixed, targeting costs scale the dispersion of log ratios of contact probabilities across individuals of the same observed type. Differences between quantiles remove the common contribution of preferences and social circles. Comparing these observed quantile differences with the corresponding differences implied by the specified payoff distribution and the probabilities of successful contact identifies the targeting-cost parameter. Once targeting costs are known, I recover utilities from contact probabilities after accounting for social-circle composition and the probability of successful contact. Utilities are identified up to the payoff from one reference partner type.

Second, suppose individual contact probabilities are observed but social circles are not. A tendency to pursue a particular type could reflect either a preference for that type or its prevalence in routine contacts. I separate these explanations by comparing environments in which targeting costs change while preferences and social circles remain fixed. An introduction service that reduces the effort required to reach partners, without changing existing networks or match payoffs, illustrates this variation. The same quantile comparisons recover targeting costs in each environment. I then identify preferences and social circles up to the reference payoff when, for each nonreference partner type, the probability of successful contact does not change in exactly the same proportion as targeting costs.

Third, suppose the researcher observes only contact probabilities averaged across individuals of each type, together with social-circle composition. These averages do not provide the distribution of individual contact probabilities needed for the quantile comparisons. I instead compare environments in which social circles change while preferences and targeting costs remain fixed. For each candidate cost and reference payoff, the observed averages imply a vector of utilities in each environment. The correct cost must reconcile these utilities across environments. Holding the reference payoff fixed, I establish local identification when, at the true cost, at least one recovered utility changes at different rates across the two environments as the candidate cost varies. Global identification holds when only one admissible cost reconciles all recovered utilities. These results identify the cost and utilities up to the reference payoff.

The analysis uses population contact probabilities and recovers the probability of successful contact from the other side's search behavior and known meeting probabilities. Section 5 states the precise data requirements and identification conditions.

Related literature. The model combines costly targeting with a social-circle interpretation of baseline search. In Cheremukhin et al. (2020), an individual's return to targeting a prospective partner depends on both the match payoff and reciprocal interest. I retain that interaction and study identification with unobserved payoff heterogeneity and nonuniform baseline opportunities. The cost specification also relates to Cheremukhin et al. (2015), who study probabilistic choice with costs measured by departures from a reference distribution. Here the reference distribution describes social contacts, and the cost represents the effort required to redirect search. The social-circle mechanism provides another explanation for sorting alongside the role of match payoffs studied by Becker (1973) and Legros and Newman (2002).

For identification, Choo and Siow (2006) and Galichon and Salanié (2022) recover match surplus in transferable-utility models using observed matching patterns and restrictions on unobserved preferences. The present analysis uses contact probabilities in a search equilibrium with given match payoffs. Its individual-data argument is a location-scale application of the monotonicity and quantile logic in Matzkin (2003). Its aggregate argument uses the convex-conjugate inversion of choice probabilities in Galichon and Salanié (2022), then compares the utilities recovered across environments to identify targeting costs.

Section 2 presents payoffs, social circles, and the meeting technology. Section 3 defines equilibrium, and Section 4 characterizes search, establishes existence, and explains assortative matching. Section 5 develops the identification results, and Section 6 concludes. Appendix A contains the proofs.

2 Model Setup

The market contains a continuum of men, $\mathcal{I}$, and a continuum of women, $\mathcal{J}$. Each man i has an observed type $x_i \in \mathcal{X}$, and each woman j has an observed type $y_j \in \mathcal{Y}$. Types describe characteristics such as education. Both type sets are finite and contain at least two elements. Let $N_x > 0$ and $M_y > 0$ denote the population masses of male type x and female type y. An individual is indifferent between prospective partners of the same type, but individuals of the same type may value those partners differently.

Index singlehood by 0, and write $\mathcal{X}_0 = \mathcal{X} \cup \{0\}$ and $\mathcal{Y}_0 = \mathcal{Y} \cup \{0\}$. Match payoffs have a component shared by individuals of the same observed type and an individual-specific component.

Assumption 2.1 (Payoffs and heterogeneity). *(i) When man i of type x matches with woman j of type y, their payoffs are $U_{xy} + \varepsilon_{iy}$ and $V_{xy} + \eta_{xj}$, respectively. The arrays (U_{xy}) and (V_{xy}) are finite and exogenous. Their singlehood payoffs are ε_{i0} and η_{0j}.*

(ii) Conditional on $x_i = x$, the vector $\boldsymbol{\varepsilon}_i = (\varepsilon_{iy})_{y \in \mathcal{Y}_0}$ has distribution μ_x. Conditional on $y_j = y$, the vector $\boldsymbol{\eta}_j = (\eta_{xj})_{x \in \mathcal{X}_0}$ has distribution ν_y. These distributions are absolutely continuous with respect to Lebesgue measure.

(iii) The payoff shocks satisfy

$$\mathbb{E}_{\mu_x}\left[\max_{y \in \mathcal{Y}_0} |\varepsilon_y|\right] < \infty, \qquad \mathbb{E}_{\nu_y}\left[\max_{x \in \mathcal{X}_0} |\eta_x|\right] < \infty.$$

Individuals observe their own payoff shocks before choosing where to search. They know the type-specific distributions of others' shocks, but not the realization of a particular prospective partner. For example, two university graduates may place different values on a partner's education because they place different weight on shared professional interests. The researcher observes their education but not these individual valuations. The analysis takes (U_{xy}) and (V_{xy}) as given and characterizes a partial equilibrium in search decisions.

A man of type x with shocks $\boldsymbol{\varepsilon}$ chooses a vector $\boldsymbol{p}_{x\cdot}(\boldsymbol{\varepsilon}) \in \Delta(\mathcal{Y})$, where p_{xy} is his probability of contacting a woman of type y and $\Delta(\mathcal{Y})$ is the probability simplex over female types. A woman of type y similarly chooses $\boldsymbol{q}_{\cdot y}(\boldsymbol{\eta}) \in \Delta(\mathcal{X})$. These probabilities describe whom agents contact; forming a match also requires reciprocal targeting and a successful meeting. Within a contacted type, the prospective partner's shocks follow that type's population distribution.

The baseline probabilities $\tilde{\boldsymbol{p}}_{x\cdot}$ and $\tilde{\boldsymbol{q}}_{\cdot y}$ describe social-circle composition and are common within each observed type. In particular, $\tilde{p}_{xy}$ is the proportion of type-y women in the social circle of type-x men. It is the probability of contacting type y without additional targeting effort. The analogous probability for a type-y woman is $\tilde{q}_{xy}$.

To describe reciprocal search, define the average targeting probabilities

$$\bar{p}_{xy} = \int p_{xy}(\boldsymbol{\varepsilon})\, d\mu_x(\boldsymbol{\varepsilon}), \qquad \bar{q}_{xy} = \int q_{xy}(\boldsymbol{\eta})\, d\nu_y(\boldsymbol{\eta}).$$

Let $r_y(x)$ be the meeting probability for a type-x man contacting type y, conditional on reciprocal targeting, and let $r_x(y)$ be its counterpart for a type-y woman contacting type x. Conditional on contacting type y, a type-x man forms a match with probability Q_{xy}. Conditional on contacting type x, a type-y woman forms a match with probability P_{xy}, where

$$Q_{xy} = r_y(x)\bar{q}_{xy}, \qquad P_{xy} = r_x(y)\bar{p}_{xy}. \tag{2.1}$$

Thus Q_{xy} and P_{xy} combine reciprocal targeting with the meeting technology.

Assumption 2.2 (Meeting technology)**.** *The probabilities $r_y(x), r_x(y) \in (0,1]$ are known to agents and the researcher. For every pair (x,y), they satisfy*

$$N_x r_y(x) = M_y r_x(y). \tag{2.2}$$

Condition (2.2) ensures that both sides count the same matches. The mass of type-(x,y) matches calculated from men is $N_x \bar{p}_{xy} Q_{xy}$; calculated from women, it is $M_y \bar{q}_{xy} P_{xy}$. Equation (2.1) makes these expressions equal under (2.2).

Departures from baseline search incur a targeting cost. To write costs additively in utility, let $c_x \leq 0$ denote the utility deduction for men, so that $-c_x$ is the cost itself. Define c_y analogously for women.

Assumption 2.3 (Targeting costs)**.** *The baseline vectors are probability vectors with strictly positive entries. For each type, the utility deductions are*

$$c_x(\boldsymbol{p}_{x\cdot}; \tilde{\boldsymbol{p}}_{x\cdot}) = -k_x \sum_{y \in \mathcal{Y}} p_{xy} \log \frac{p_{xy}}{\tilde{p}_{xy}}, \qquad k_x > 0, \tag{2.3}$$

$$c_y(\boldsymbol{q}_{\cdot y}; \tilde{\boldsymbol{q}}_{\cdot y}) = -k_y \sum_{x \in \mathcal{X}} q_{xy} \log \frac{q_{xy}}{\tilde{q}_{xy}}, \qquad k_y > 0. \tag{2.4}$$

The convention is $0 \log(0/a) = 0$ for $a > 0$. Zero and infinite values of the cost parameters k_x and k_y are considered only as limits.

The sums in (2.3) and (2.4) are relative entropies; see Cover and Thomas (2006). They measure departures from the baseline probabilities, as in the probabilistic-choice framework of Cheremukhin et al. (2015). Targeting costs are nonnegative and equal zero only at the baseline. Concentrating all search on type y costs $k_x \log(1/\tilde{p}_{xy})$, so exclusive targeting is more costly when that type is rarer in the social circle. The parameters k_x and k_y scale costs relative to match payoffs.

For example, suppose graduates make up 70 percent of a person's social circle. With two partner types and $k_x = 1$, raising the probability of contacting a graduate to 90 percent costs

$$0.9 \log \frac{0.9}{0.7} + 0.1 \log \frac{0.1}{0.3} \simeq 0.116$$

utility units.

Given the other side's search decisions, a searching man's expected utility equals his singlehood payoff plus expected gains from matching, less the targeting cost:

$$u_x(\boldsymbol{p}_{x\cdot};\boldsymbol{Q}_{x\cdot},\boldsymbol{U}_{x\cdot},\boldsymbol{\varepsilon}) = \varepsilon_0 + \sum_{y\in\mathcal{Y}} p_{xy}Q_{xy}(U_{xy}+\varepsilon_y-\varepsilon_0) + c_x(\boldsymbol{p}_{x\cdot};\tilde{\boldsymbol{p}}_{x\cdot}). \tag{2.5}$$

Here $p_{xy}Q_{xy}$ is the probability of matching with type y, and $1-\sum_y p_{xy}Q_{xy}$ is the probability of remaining single after search. A searching woman's expected utility is

$$v_y(\boldsymbol{q}_{\cdot y};\boldsymbol{P}_{\cdot y},\boldsymbol{V}_{\cdot y},\boldsymbol{\eta}) = \eta_0 + \sum_{x\in\mathcal{X}} q_{xy}P_{xy}(V_{xy}+\eta_x-\eta_0) + c_y(\boldsymbol{q}_{\cdot y};\tilde{\boldsymbol{q}}_{\cdot y}). \tag{2.6}$$

3 Targeted-Search Equilibrium

The paper considers markets in which everyone participates in search. The following restriction makes every potential match at least as valuable as singlehood before targeting costs.

Assumption 3.1 (No voluntary singlehood)**.** *For every male type x and female type y,*

$$\varepsilon_0 \le \min_{z\in\mathcal{Y}}(U_{xz}+\varepsilon_z) \quad \mu_x\text{-almost surely}, \qquad \eta_0 \le \min_{z\in\mathcal{X}}(V_{zy}+\eta_z) \quad \nu_y\text{-almost surely}.$$

Absolute continuity in Assumption 2.1 makes ties between any match and singlehood a probability-zero event. Baseline search has zero cost under Assumption 2.3. Once equilibrium probabilities of successful contact are positive, baseline search therefore yields a strictly positive expected gain over singlehood. Individuals may remain single because search fails, but they prefer to participate. Proposition 4.2 establishes the required positivity.

Let $\mathcal{E}_x$ and $\mathcal{H}_y$ denote the supports of μ_x and ν_y. An equilibrium requires each individual's search decision to be optimal and the probabilities $\boldsymbol{P}$ and $\boldsymbol{Q}$ to agree with the decisions on the other side.

Definition 3.1. *Given the payoff arrays, social-circle baselines, cost parameters, shock distributions, and meeting technology, a* targeted-search equilibrium *consists of measurable rules*

$$\boldsymbol{p}^*_{x\cdot} : \mathcal{E}_x \longrightarrow \Delta(\mathcal{Y}), \qquad \boldsymbol{q}^*_{\cdot y} : \mathcal{H}_y \longrightarrow \Delta(\mathcal{X})$$

for every x and y. For almost every shock realization, these rules satisfy

$$\boldsymbol{p}^*_{x\cdot}(\boldsymbol{\varepsilon}) \in \arg\max_{\boldsymbol{p}\in\Delta(\mathcal{Y})} u_x(\boldsymbol{p};\boldsymbol{Q}^*_{x\cdot},\boldsymbol{U}_{x\cdot},\boldsymbol{\varepsilon}), \tag{3.1}$$

$$\boldsymbol{q}^*_{\cdot y}(\boldsymbol{\eta}) \in \arg\max_{\boldsymbol{q}\in\Delta(\mathcal{X})} v_y(\boldsymbol{q};\boldsymbol{P}^*_{\cdot y},\boldsymbol{V}_{\cdot y},\boldsymbol{\eta}), \tag{3.2}$$

where

$$P^*_{xy} = r_x(y) \int p^*_{xy}(\boldsymbol{\varepsilon})\, d\mu_x(\boldsymbol{\varepsilon}), \tag{3.3a}$$

$$Q^*_{xy} = r_y(x) \int q^*_{xy}(\boldsymbol{\eta})\, d\nu_y(\boldsymbol{\eta}). \tag{3.3b}$$

Men respond to women's average targeting through $\boldsymbol{Q}^*$, and women respond to men's average targeting through $\boldsymbol{P}^*$. The equilibrium is partial because the payoff arrays remain fixed throughout this adjustment; the model does not determine transfers or how match payoffs are divided.

4 Characterization of Equilibrium

Relative-entropy costs make each individual's search problem strictly concave. This section derives the unique best response to any given probabilities of successful contact, establishes equilibrium existence, and explains the role of social circles in assortative matching.

4.1 Optimal Targeting

Fix a man's type, payoff shocks, and vector $\boldsymbol{Q}_{x\cdot}$. The expected gain from contacting type y is $Q_{xy}(U_{xy} + \varepsilon_y - \varepsilon_0)$. The individual trades this gain against the cost of departing from his social-circle baseline.

Proposition 4.1 (Best responses)**.** *Under Assumptions 2.1 and 2.3, the unique maximizers of (2.5) and (2.6) over their respective simplexes, for any $\boldsymbol{Q}_{x\cdot} \in [0,1]^{|\mathcal{Y}|}$ and $\boldsymbol{P}_{\cdot y} \in [0,1]^{|\mathcal{X}|}$ are*

$$p_{xy}(\boldsymbol{\varepsilon}; \boldsymbol{Q}_{x\cdot}) = \frac{\tilde{p}_{xy} \exp\{Q_{xy}(U_{xy} + \varepsilon_y - \varepsilon_0)/k_x\}}{\sum_{z\in\mathcal{Y}} \tilde{p}_{xz} \exp\{Q_{xz}(U_{xz} + \varepsilon_z - \varepsilon_0)/k_x\}}, \tag{4.1a}$$

$$q_{xy}(\boldsymbol{\eta}; \boldsymbol{P}_{\cdot y}) = \frac{\tilde{q}_{xy} \exp\{P_{xy}(V_{xy} + \eta_x - \eta_0)/k_y\}}{\sum_{z\in\mathcal{X}} \tilde{q}_{zy} \exp\{P_{zy}(V_{zy} + \eta_z - \eta_0)/k_y\}}. \tag{4.1b}$$

All components are strictly positive. In equilibrium, $p^*_{xy}(\boldsymbol{\varepsilon}) = p_{xy}(\boldsymbol{\varepsilon}; \boldsymbol{Q}^*_{x\cdot})$ and $q^*_{xy}(\boldsymbol{\eta}) = q_{xy}(\boldsymbol{\eta}; \boldsymbol{P}^*_{\cdot y})$ almost surely under the respective shock laws.

The best response has logit form conditional on payoff shocks. A type receives more search when its expected match gain is larger, holding the other terms fixed. Its baseline share also matters: a type that is common in the social circle is less costly to reach. These are properties of the individual best response; equilibrium changes also reflect the other side's adjustment.

The ratio of two targeting probabilities makes the three determinants explicit:

$$\log \frac{p^*_{xy}}{p^*_{xy'}} = \log \frac{\tilde{p}_{xy}}{\tilde{p}_{xy'}} + \frac{Q^*_{xy}(U_{xy} + \varepsilon_y - \varepsilon_0) - Q^*_{xy'}(U_{xy'} + \varepsilon_{y'} - \varepsilon_0)}{k_x}. \tag{4.2}$$

Relative contact frequencies depend on relative social-circle shares and differences in expected match gains, scaled by targeting costs. Equation (4.2) is the basis of the identification analysis.

Corollary 4.1 (High targeting costs)**.** *Under the assumptions of Proposition 4.1, fix payoffs, shock distributions, and social-circle baselines. Along any sequence of equilibria, for almost every shock realization,*

$$p^*_{xy}(\boldsymbol{\varepsilon}) \longrightarrow \tilde{p}_{xy} \quad \text{as } k_x \longrightarrow \infty, \qquad q^*_{xy}(\boldsymbol{\eta}) \longrightarrow \tilde{q}_{xy} \quad \text{as } k_y \longrightarrow \infty.$$

Reciprocation and meeting probabilities are bounded by one. For fixed payoff shocks, the exponents in (4.1) therefore converge to zero as the corresponding cost parameter grows. Search converges to the social-circle baseline even if the other side's decisions vary along the sequence. Since targeting probabilities are bounded by one, their population averages converge to the same baseline.

Corollary 4.2 (Low targeting costs)**.** *Under the assumptions of Proposition 4.1, fix payoffs, shock distributions, and social-circle baselines. Consider equilibria indexed by n, with $k_x^{(n)} \to 0$ and $Q_{xy}^{*(n)} \to \bar{Q}_{xy}$ for every y. For almost every shock realization, if the maximizer*

$$y^\dagger = \arg\max_{y \in \mathcal{Y}} \bar{Q}_{xy}(U_{xy} + \varepsilon_y - \varepsilon_0)$$

is unique, then $p_{xy^\dagger}^{*(n)} \to 1$. The corresponding statement holds for women when $k_y^{(n)} \to 0$ and $P_{xy}^{*(n)}$ converges for every x.

With low targeting costs, search concentrates on the largest expected gain, which incorporates the chance of reciprocal interest as well as the value of a match.

4.2 Equilibrium Existence

Although best responses are decision rules for a continuum of individuals, they interact only through the finite arrays $\boldsymbol{P}$ and $\boldsymbol{Q}$. Substituting Proposition 4.1 into (3.3) gives the equilibrium conditions

$$P^*_{xy} = r_x(y) \int p_{xy}(\boldsymbol{\varepsilon}; \boldsymbol{Q}^*_{x\cdot})\, d\mu_x(\boldsymbol{\varepsilon}), \tag{4.3a}$$

$$Q^*_{xy} = r_y(x) \int q_{xy}(\boldsymbol{\eta}; \boldsymbol{P}^*_{\cdot y})\, d\nu_y(\boldsymbol{\eta}). \tag{4.3b}$$

The right-hand side is continuous in $(\boldsymbol{P}^*, \boldsymbol{Q}^*)$ and takes the compact, convex cube $[0,1]^{2|\mathcal{X}||\mathcal{Y}|}$ into itself. Brouwer's fixed-point theorem gives the following result.

Proposition 4.2 (Equilibrium existence)**.** *Under Assumptions 2.1, 2.2, 2.3, and 3.1, a targeted-search equilibrium exists. Every equilibrium has $P^*_{xy} > 0$ and $Q^*_{xy} > 0$ for all (x, y), and almost every individual strictly prefers search to singlehood.*

Appendix A verifies continuity, participation, and consistency of the implied match masses. The result establishes existence; the identification analysis conditions on the observed equilibrium in each market.

4.3 Social Circles and Assortative Matching

Positive assortative matching means that similar types tend to match with one another. Corollary 4.1 shows how social circles can generate this pattern. If targeting costs grow on both sides, the probability that a type-x man matches with type y converges to

$$\tilde{p}^*_{xy} Q^*_{xy} \longrightarrow r_y(x) \tilde{p}_{xy} \tilde{q}_{xy}. \tag{4.4}$$

Thus matching depends on the overlap of the two sides' baseline opportunities. A social circle concentrated on similar types favors similar partners when that concentration is reciprocated.

For an exact illustration, consider two ordered types on each side, with equal population masses and meeting probabilities equal to one. Suppose every type's social circle assigns probability $a \in (1/2, 1)$ to its own type and $1-a$ to the other type. In the high-cost limit, the probabilities of a same-type match and a different-type match are a^2 and $(1-a)^2$, respectively. Conditional on forming a match, the probability of a same-type partner is therefore

$$\frac{a^2}{a^2 + (1-a)^2} > \frac{1}{2}.$$

For $a = 0.7$, this probability is approximately 0.845, compared with 0.5 when both baselines are uniform. Match payoffs do not enter this limit, so the same sorting arises when systematic utilities are constant across partner types.

5 Identification

The objective is to recover match utilities and targeting costs, and to recover social-circle composition when it is unobserved. Fix a male type x throughout this section; the same arguments apply to women. An *environment* is a market or policy setting in which the model has an observed equilibrium. Equilibrium asterisks are suppressed in this section.

The data concern search decisions. With individual data, the researcher observes the population distribution of the vector $\boldsymbol{p}_{x\cdot}(\boldsymbol{\varepsilon})$ across type-x men. These are individual contact probabilities, not a single realized contact per person. With aggregate data, the researcher observes only the averages $\boldsymbol{m}_x = (m_{xy})_{y\in\mathcal{Y}}$, where $m_{xy} = \mathbb{E}_{\mu_x}[p_{xy}(\boldsymbol{\varepsilon})]$. In either case, probabilities of successful contact are recovered from the other side's average targeting and the meeting technology.

Assumption 5.1 (Information for identification)**.** *In each observed environment, search decisions arise from an equilibrium satisfying Assumptions 2.1, 2.2, 2.3, and 3.1. The researcher knows:*

(i) the type-specific payoff-shock distribution μ_x, including its location and scale; this distribution is unchanged across the environments being compared;

(ii) women's average targeting probabilities $\bar{q}_{xy}$ and meeting probabilities $r_y(x)$ for every y, and hence $Q_{xy} = r_y(x)\bar{q}_{xy} > 0$ in each environment.

For utility recovery, choose a reference partner type y^1. The results identify parameters *up to* U_{xy^1}: they determine the remaining parameters as functions of this reference payoff, with local or global uniqueness as stated. Uniqueness is assessed with the reference value held fixed; identification of the reference payoff itself is left open. Even when circles and costs are known, search ratios in one environment determine differences in $Q_{xy}U_{xy}$ rather than their common level. Adding the same constant to every $Q_{xy}U_{xy}$ leaves targeting probabilities unchanged whenever the resulting utilities remain admissible. The reference payoff selects one member of this observationally equivalent family. Candidate parameters must satisfy the model assumptions, including the participation restriction, under the specified shock law.

5.1 Individual Targeting with Observed Social Circles

For any $y \neq y^1$, define the log ratio of contact probabilities and the corresponding payoff-shock difference by

$$\beta_y = \log \frac{p_{xy}(\boldsymbol{\varepsilon})}{p_{xy^1}(\boldsymbol{\varepsilon})},$$
$$\tilde{\epsilon}_y = Q_{xy}(\varepsilon_y - \varepsilon_0) - Q_{xy^1}(\varepsilon_{y^1} - \varepsilon_0).$$

The distribution of β_y is observed in individual data. The distribution of $\tilde{\epsilon}_y$ is known from μ_x and $\boldsymbol{Q}_{x\cdot}$. With

$$a_y = \log \frac{\tilde{p}_{xy}}{\tilde{p}_{xy^1}}, \qquad d_y = Q_{xy}U_{xy} - Q_{xy^1}U_{xy^1},$$

the optimality condition (4.2) becomes

$$k_x \beta_y = k_x a_y + d_y + \tilde{\epsilon}_y. \tag{5.1}$$

Holding $\boldsymbol{Q}_{x\cdot}$ fixed, social circles and systematic utilities shift the location of β_y, whereas targeting costs scale its variation across individuals of the same observed type.

Let $F^{-1}(u) = \inf\{z : F(z) \geq u\}$, for $0 < u < 1$, denote a generalized quantile. Write $b_y(u)$ for the u-quantile of β_y and $e_y(u)$ for the u-quantile of $\tilde{\epsilon}_y$, both conditional on type x.

Proposition 5.1 (Targeting costs from individual probabilities)**.** *Under Assumption 5.1, the population distribution of β_y for any $y \neq y^1$ identifies k_x. For $0 < u < v < 1$ with $b_y(v) > b_y(u)$,*

$$k_x = \frac{e_y(v) - e_y(u)}{b_y(v) - b_y(u)}. \tag{5.2}$$

Such quantiles exist. Social-circle composition need not be observed for this result.

The difference between two quantiles removes the common term $k_x a_y + d_y$ in (5.1). Absolute continuity and positive Q_{xy} make the scalar shock $\tilde{\epsilon}_y$ nondegenerate.

Once k_x is identified, any quantile identifies the intercept

$$\ell_y = k_x b_y(u) - e_y(u) = k_x a_y + d_y. \tag{5.3}$$

Under the model, ℓ_y is the same for every $u \in (0,1)$. Observed social circles separate the baseline contribution $k_x a_y$ from the utility difference d_y.

Proposition 5.2 (Utilities with observed social circles)**.** *Under Assumption 5.1, suppose the population distribution of individual targeting probabilities and the baseline $\tilde{\boldsymbol{p}}_{x\cdot}$ are observed. The utility vector is identified up to U_{xy^1} through*

$$U_{xy} = \frac{\ell_y - k_x a_y + Q_{xy^1} U_{xy^1}}{Q_{xy}}, \qquad y \neq y^1. \tag{5.4}$$

Equation (5.4) adjusts relative search for both social-circle composition and the probability that contact results in a match. It therefore distinguishes a type's attractiveness from its availability through routine contacts.

5.2 Individual Targeting with Unobserved Social Circles

When social circles are unobserved, the intercept ℓ_y combines baseline opportunities and preferences. A second environment can separate them if preferences and social circles remain fixed while targeting costs change. An introduction service that reduces the effort required to reach prospective partners provides one possible interpretation of this variation. The identifying restriction is that the service changes neither match payoffs nor the baseline composition of routine contacts. Contact-success probabilities may adjust and are observed in both environments.

Index the two environments by $t = 1, 2$. Proposition 5.1 recovers each $k_{x,t}$, and (5.3) recovers each ℓ_{yt} using that environment's contact-probability distribution and $\boldsymbol{Q}_{x\cdot,t}$.

Proposition 5.3 (Utilities and unobserved social circles)**.** *Under Assumption 5.1, suppose the population distribution of $\boldsymbol{p}_{x\cdot}(\boldsymbol{\varepsilon})$ is observed in each of two environments with common utilities $\boldsymbol{U}_{x\cdot}$ and baseline $\tilde{\boldsymbol{p}}_{x\cdot}$. If*

$$k_{x,1} Q_{xy,2} - k_{x,2} Q_{xy,1} \neq 0 \qquad \text{for every } y \neq y^1, \tag{5.5}$$

then $\boldsymbol{U}_{x\cdot}$ and $\tilde{\boldsymbol{p}}_{x\cdot}$ are identified up to U_{xy^1}.

For each nonreference type, the two environments provide two linear equations in its baseline log ratio a_y and utility U_{xy}:

$$k_{x,t} a_y + Q_{xy,t} U_{xy} = \ell_{yt} + Q_{xy^1,t} U_{xy^1}, \qquad t = 1, 2. \tag{5.6}$$

Condition (5.5) makes this system invertible. Equivalently, $Q_{xy,t}/k_{x,t}$ must differ across the two environments: the probability of successful contact with type y cannot change in exactly the same proportion as targeting costs. If $Q_{xy,t}$ is unchanged, any change in $k_{x,t}$ satisfies the condition. After recovering the log ratios, the requirement that baseline probabilities sum to one recovers their levels.

5.3 Aggregate Targeting Shares

Now suppose individual contact probabilities are unobserved. The researcher observes only their averages m_{xy} and the social-circle baseline. These *aggregate targeting shares* describe contacts, rather than realized matches. Averaging the best response gives

$$m_{xy} = \int \frac{\tilde{p}_{xy} \exp\{Q_{xy}(U_{xy} + \varepsilon_y - \varepsilon_0)/k_x\}}{\sum_{z \in \mathcal{Y}} \tilde{p}_{xz} \exp\{Q_{xz}(U_{xz} + \varepsilon_z - \varepsilon_0)/k_x\}} \, d\mu_x(\boldsymbol{\varepsilon}). \tag{5.7}$$

Each share is strictly positive. These averages do not reveal the distribution of β_y used in Proposition 5.1.

For a given targeting cost, the aggregate shares determine differences in the systematic returns to targeting each partner type. These returns combine match payoffs and social-circle opportunities. For a candidate cost $\kappa > 0$, let $g_y^*(\boldsymbol{m}_x; \boldsymbol{Q}_{x\cdot}, \kappa)$ denote the difference relative to type y^1, expressed in payoff units, that reproduces the observed shares. The reference component is $g_{y^1}^* = 0$. The proof of Proposition 5.4 in Appendix A constructs this inverse and establishes its existence, uniqueness, and differentiability in the candidate cost. At the true cost,

$$g_y^*(\boldsymbol{m}_x; \boldsymbol{Q}_{x\cdot}, k_x) = k_x \log \frac{\tilde{p}_{xy}}{\tilde{p}_{xy^1}} + Q_{xy} U_{xy} - Q_{xy^1} U_{xy^1}. \tag{5.8}$$

For a given reference payoff, one environment supplies a vector of implied utilities for each candidate cost. To distinguish the costs, compare two environments with different observed social circles but common utilities and a common targeting-cost parameter. A change in the educational composition of routine contacts is one possible source of variation, provided it leaves preferences, payoff heterogeneity, and the cost of redirecting search unchanged. The inverse must use each environment's own observed probabilities of successful contact.

For $t = 1, 2$ and $y \neq y^1$, let $a_{yt} = \log(\tilde{p}_{xy,t}/\tilde{p}_{xy^1,t})$. At a candidate cost κ and reference payoff U_{xy^1}, environment t implies the utility

$$U_{xy,t}(\kappa) = \frac{g_y^*(\boldsymbol{m}_{x,t}; \boldsymbol{Q}_{x\cdot,t}, \kappa) - \kappa a_{yt} + Q_{xy^1,t} U_{xy^1}}{Q_{xy,t}}. \tag{5.9}$$

The reference payoff is held fixed in these functions and their derivatives with respect to κ. A cost is consistent with common preferences only if it yields the same utility in both environments. Define the difference

$$H_y(\kappa) = U_{xy,1}(\kappa) - U_{xy,2}(\kappa), \qquad y \neq y^1. \tag{5.10}$$

Proposition 5.4 (Identification from aggregate shares)**.** *Under Assumption 5.1, suppose aggregate targeting shares and social-circle baselines are observed in two environments with common utilities and cost k_x. Let H_y be defined by (5.10).*

(i) If $H_y'(k_x) \neq 0$ for at least one nonreference type y, then k_x and the utilities are locally identified up to U_{xy^1}.

(ii) If k_x is the unique positive solution of $H_y(\kappa) = 0$ for every $y \neq y^1$ whose implied utilities are admissible in both environments, then k_x and the utilities are globally identified up to U_{xy^1}.

The local condition requires a nonzero first-order change in at least one utility difference when the candidate cost changes. It is equivalent to full column rank of the joint restrictions in the cost and nonreference utilities; the appendix shows this equivalence. The global condition rules out a second admissible cost that also reconciles all recovered utilities.

6 Concluding Remarks

Matching among similar people can reflect social-circle composition as well as preferences. Social circles determine whom people contact without additional effort, and targeting costs govern departures from those opportunities. Reciprocal search translates contact decisions into matches. The high-cost limit makes this mechanism explicit: matching probabilities depend on the overlap of the two sides' baseline contacts, even when systematic payoffs do not favor similar partners.

The identification results show how contact probabilities separate preferences from social-circle opportunities. Quantile comparisons using a specified distribution of unobserved payoff heterogeneity identify targeting costs from individual contact probabilities. Observed circles then identify preferences, while changes in costs recover preferences and circles when circles are unobserved. With aggregate shares, identification instead uses changes in circles that leave preferences and costs fixed. The remaining parameters are identified up to one reference payoff, under the stated local or global conditions. Search data thus distinguish whom people prefer from whom they can readily meet.

A Proofs

Proof of Proposition 4.1. Fix a male type x, a finite payoff-shock vector $\boldsymbol{\varepsilon}$, and $\boldsymbol{Q}_{x\cdot} \in [0,1]^{|\mathcal{Y}|}$. Write $A_y = Q_{xy}(U_{xy} + \varepsilon_y - \varepsilon_0)$. Dropping the constant singlehood payoff, the problem is

$$\max_{\boldsymbol{p} \in \Delta(\mathcal{Y})} \left\{ \sum_y p_y A_y - k_x \sum_y p_y \log \frac{p_y}{\tilde{p}_{xy}} \right\}.$$

The objective is continuous on the compact simplex because $t \log t \to 0$ as $t \downarrow 0$. It is strictly concave because $k_x > 0$ and $-\sum_y p_y \log p_y$ is strictly concave. A unique maximizer therefore exists.

The maximizer is interior. If $p_y = 0$, choose a coordinate z with $p_z > 0$ and transfer mass $h \in (0, p_z)$ from z to y. The change in the objective is $-k_x h \log h + O(h)$ as $h \downarrow 0$, which is positive for sufficiently small h. A boundary point cannot maximize the objective.

The first-order condition for each y, with multiplier λ on $\sum_y p_y = 1$, is

$$A_y - k_x \left(\log \frac{p_y}{\tilde{p}_{xy}} + 1 \right) = \lambda.$$

Consequently $p_y = \tilde{p}_{xy} \exp(A_y/k_x) \exp(-1 - \lambda/k_x)$. Summing over y determines the common factor and gives (4.1a). Strict concavity makes these conditions sufficient. The same argument, with V, P, $\boldsymbol{\eta}$, and $\tilde{\boldsymbol{q}}$, proves (4.1b). Each numerator and denominator is strictly positive, so every best-response probability is positive. □

Proof of Corollaries 4.1 and 4.2. For either sequence of equilibria, the best-response formulas hold simultaneously for all n on sets of full measure under the respective shock laws, since the sequence is countable. Work on these sets throughout.

For the high-cost result, fix x and $\boldsymbol{\varepsilon}$, and let $k_x^{(n)} \to \infty$. Since $0 \le Q_{xy}^{*(n)} \le 1$,

$$\left| \frac{Q_{xy}^{*(n)}(U_{xy} + \varepsilon_y - \varepsilon_0)}{k_x^{(n)}} \right| \le \frac{|U_{xy} + \varepsilon_y - \varepsilon_0|}{k_x^{(n)}} \longrightarrow 0.$$

There are finitely many partner types. The numerator of (4.1a) converges to $\tilde{p}_{xy}$ and its denominator to $\sum_z \tilde{p}_{xz} = 1$. This proves pointwise convergence of the targeting probability. The bound $0 \le p_{xy}^{*(n)} \le 1$ permits dominated convergence under μ_x, giving $\bar{p}_{xy}^{*(n)} \to \tilde{p}_{xy}$. The argument for women is identical.

For the low-cost result, fix the sequence and shock realization in Corollary 4.2, and write $A_y^{(n)} = Q_{xy}^{*(n)}(U_{xy} + \varepsilon_y - \varepsilon_0)$. Uniqueness of the limiting maximizer implies $A_y^{(n)} - A_{y^\dagger}^{(n)} \to \Delta_y < 0$ for each $y \neq y^\dagger$. Thus

$$\frac{p_{xy}^{*(n)}}{p_{xy^\dagger}^{*(n)}} = \frac{\tilde{p}_{xy}}{\tilde{p}_{xy^\dagger}} \exp\left(\frac{A_y^{(n)} - A_{y^\dagger}^{(n)}}{k_x^{(n)}} \right) \longrightarrow 0.$$

There are finitely many nonmaximizing types, so summing these ratios gives $p_{xy^\dagger}^{*(n)} \to 1$. The women's result follows in the same way. □

Proof of Proposition 4.2. Let $\mathcal{K} = [0,1]^{2|\mathcal{X}||\mathcal{Y}|}$. For $(\boldsymbol{P}, \boldsymbol{Q}) \in \mathcal{K}$, define the two blocks of a map $T : \mathcal{K} \to \mathcal{K}$ by

$$T_{xy}^P(\boldsymbol{P}, \boldsymbol{Q}) = r_x(y) \int p_{xy}(\boldsymbol{\varepsilon}; \boldsymbol{Q}_{x\cdot})\, d\mu_x(\boldsymbol{\varepsilon}),$$
$$T_{xy}^Q(\boldsymbol{P}, \boldsymbol{Q}) = r_y(x) \int q_{xy}(\boldsymbol{\eta}; \boldsymbol{P}_{\cdot y})\, d\nu_y(\boldsymbol{\eta}).$$

The best responses in Proposition 4.1 lie in the relevant simplexes and are continuous, hence measurable, in the shock vectors. Since meeting probabilities lie in $(0,1]$, every coordinate of T lies in $[0,1]$.

To check continuity, take any sequence $(\boldsymbol{P}^{(n)}, \boldsymbol{Q}^{(n)}) \to (\boldsymbol{P}, \boldsymbol{Q})$ in $\mathcal{K}$. For every finite $\boldsymbol{\varepsilon}$, the finite sums and exponentials in (4.1a) imply

$$p_{xy}(\boldsymbol{\varepsilon}; \boldsymbol{Q}_{x\cdot}^{(n)}) \longrightarrow p_{xy}(\boldsymbol{\varepsilon}; \boldsymbol{Q}_{x\cdot}).$$

The integrands are bounded by one, an integrable bound under the probability law μ_x. Dominated convergence gives convergence of each T_{xy}^P; the same argument under ν_y gives convergence of each T_{xy}^Q. With finitely many coordinates, T is continuous.

The set $\mathcal{K}$ is nonempty, compact, and convex. Brouwer's fixed-point theorem therefore gives $(\boldsymbol{P}^*, \boldsymbol{Q}^*) = T(\boldsymbol{P}^*, \boldsymbol{Q}^*)$. The associated best responses satisfy Definition 3.1. Every best-response probability is strictly positive, so its expectation is positive; positive meeting probabilities then imply $P_{xy}^* > 0$ and $Q_{xy}^* > 0$.

For participation, Assumption 3.1 gives $U_{xy} + \varepsilon_y - \varepsilon_0 \geq 0$ almost surely. For a fixed y, equality describes a hyperplane in the shock vector and has probability zero by Assumption 2.1. There are finitely many types, so all these gains are strictly positive simultaneously almost surely. At baseline search, the expected gain over singlehood is

$$u_x(\tilde{\boldsymbol{p}}_{x\cdot}; \boldsymbol{Q}_{x\cdot}^*, \boldsymbol{U}_{x\cdot}, \boldsymbol{\varepsilon}) - \varepsilon_0 = \sum_y \tilde{p}_{xy} Q_{xy}^*(U_{xy} + \varepsilon_y - \varepsilon_0) > 0.$$

Optimal search does at least as well. The same argument applies to women.

Finally, define the implied match mass

$$\lambda_{xy} = N_x \bar{p}^*_{xy} Q^*_{xy} = N_x r_y(x) \bar{p}^*_{xy} \bar{q}^*_{xy} = M_y \bar{q}^*_{xy} P^*_{xy},$$

where the last equality uses Assumption 2.2. Since $P^*, Q^* \leq 1$ and targeting probabilities sum to one, $\sum_y \lambda_{xy} \leq N_x$ and $\sum_x \lambda_{xy} \leq M_y$. Both sides therefore imply the same nonnegative match masses, with the remaining mass single. □

Proof of Proposition 5.1. Fix x, a nonreference type y, and the observed equilibrium vector $\boldsymbol{Q}_{x\cdot}$. The variable

$$\tilde{\epsilon}_y = Q_{xy}\varepsilon_y - Q_{xy^1}\varepsilon_{y^1} + (Q_{xy^1} - Q_{xy})\varepsilon_0$$

is a nonzero linear projection of $\boldsymbol{\varepsilon}$: its coefficient on ε_y is $Q_{xy} > 0$. An absolutely continuous vector has an absolutely continuous distribution under a nonzero scalar projection. Hence $\tilde{\epsilon}_y$ has a continuous, nondegenerate distribution. Its law is known from Assumption 5.1.

Equation (5.1) is an increasing affine transformation because $k_x > 0$. In particular,

$$F_{\beta_y|x}(b) = F_{\tilde{\epsilon}_y|x}\big(k_x b - k_x a_y - d_y\big).$$

Increasing affine transformations commute with generalized quantiles, so for every $u \in (0,1)$,

$$k_x b_y(u) = k_x a_y + d_y + e_y(u). \tag{A.1}$$

The use of generalized quantiles permits gaps in the support. Nondegeneracy gives $0 < u < v < 1$ with $b_y(v) > b_y(u)$. Subtracting (A.1) at u from the same identity at v gives (5.2). Both quantile functions are known, so any parameter vector producing the observed distribution must have this value of k_x. □

Proof of Proposition 5.2. Proposition 5.1 identifies k_x. Equation (A.1) then identifies $\ell_y = k_x b_y(u) - e_y(u)$ for each $y \neq y^1$, independently of the chosen $u \in (0,1)$. Observed baseline probabilities identify a_y. Substituting $d_y = Q_{xy}U_{xy} - Q_{xy^1}U_{xy^1}$ into $\ell_y = k_x a_y + d_y$ gives (5.4). The denominator Q_{xy} is strictly positive. Hence each nonreference utility is a uniquely determined function of U_{xy^1}, proving identification up to this payoff. □

Proof of Proposition 5.3. Fix the reference payoff U_{xy^1}. In each environment t, Proposition 5.1 identifies $k_{x,t}$ and (5.3) identifies ℓ_{yt}. Because preferences and social circles are unchanged, the same unknown pair (a_y, U_{xy}) satisfies

$$\begin{pmatrix} k_{x,1} & Q_{xy,1} \\ k_{x,2} & Q_{xy,2} \end{pmatrix} \begin{pmatrix} a_y \\ U_{xy} \end{pmatrix} = \begin{pmatrix} \ell_{y1} + Q_{xy^1,1} U_{xy^1} \\ \ell_{y2} + Q_{xy^1,2} U_{xy^1} \end{pmatrix}.$$

The determinant is nonzero by (5.5), so the pair is uniquely determined. This holds for every $y \neq y^1$. Since $a_y = \log(\tilde{p}_{xy}/\tilde{p}_{xy^1})$ and the baseline probabilities sum to one,

$$\tilde{p}_{xy^1} = \frac{1}{1 + \sum_{z \neq y^1} \exp a_z}, \qquad \tilde{p}_{xy} = \frac{\exp a_y}{1 + \sum_{z \neq y^1} \exp a_z}. \qquad \square$$

Proof of Proposition 5.4. Fix the observed shares, baselines, and probabilities of successful contact in both environments. I first construct the aggregate inverse used in Section 5.3, then use it to recover the cost and utilities up to U_{xy^1}.

Step 1. Inverting aggregate targeting shares. Consider one environment and fix a candidate cost $\kappa > 0$. Set $\xi_y(\kappa) = Q_{xy}(\varepsilon_y - \varepsilon_0)/\kappa$. The systematic index in (5.7) is $Q_{xy} U_{xy}/k_x + \log \tilde{p}_{xy}$. For a candidate vector of systematic indices $\boldsymbol{\delta} \in \mathbb{R}^{|\mathcal{Y}|}$, define

$$S_y(\boldsymbol{\delta}; \mathbf{Q}_{x\cdot}, \kappa) = \int \frac{\exp\{\delta_y + \xi_y(\kappa)\}}{\sum_{z \in \mathcal{Y}} \exp\{\delta_z + \xi_z(\kappa)\}} \, d\mu_x(\boldsymbol{\varepsilon}). \tag{A.2}$$

Thus $\boldsymbol{S} : \mathbb{R}^{|\mathcal{Y}|} \to \Delta^\circ(\mathcal{Y})$, where $\Delta^\circ(\mathcal{Y})$ denotes the probability vectors with every entry strictly positive.

On the same full index space, define the expected log-sum-exp function

$$G(\boldsymbol{\delta}; \mathbf{Q}_{x\cdot}, \kappa) = \mathbb{E}_{\mu_x}\left[\log \sum_{y \in \mathcal{Y}} \exp(\delta_y + \xi_y) \right].$$

The arguments $\mathbf{Q}_{x\cdot}$ and κ are held fixed initially. Assumption 2.1 implies

$$\mathbb{E}_{\mu_x} \max_y |\xi_y| \leq \frac{2}{\kappa} \mathbb{E}_{\mu_x} \max_{z \in \mathcal{Y}_0} |\varepsilon_z| < \infty.$$

The inequalities

$$\max_y \delta_y - \max_y |\xi_y| \leq \log \sum_y \exp(\delta_y + \xi_y) \leq \max_y \delta_y + \max_y |\xi_y| + \log |\mathcal{Y}|$$

therefore show that G is finite at every $\boldsymbol{\delta}$.

Write the conditional logit probability as

$$\pi_y(\boldsymbol{\delta}, \boldsymbol{\xi}) = \frac{\exp(\delta_y + \xi_y)}{\sum_z \exp(\delta_z + \xi_z)}.$$

Its entries and first derivatives with respect to $\boldsymbol{\delta}$ are uniformly bounded. Differentiation under the expectation is consequently valid and gives

$$\nabla G = \mathbb{E}_{\mu_x}[\boldsymbol{\pi}], \qquad \nabla^2 G = \mathbb{E}_{\mu_x}\big[\operatorname{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}^\top\big].$$

These derivatives are continuous by dominated convergence. In particular, $\partial G / \partial \delta_y = S_y$.

Adding a common constant to $\boldsymbol{\delta}$ leaves $\boldsymbol{\pi}$ unchanged. Normalize $\delta_{y^1} = 0$, let $\boldsymbol{d} = (\delta_y)_{y \neq y^1}$, and write $\bar{G}(\boldsymbol{d}; \boldsymbol{Q}_{x\cdot}, \kappa) = G((0, \boldsymbol{d}); \boldsymbol{Q}_{x\cdot}, \kappa)$, with the reference coordinate first. For a nonzero vector $\boldsymbol{h}$ of nonreference coordinates, set $h_{y^1} = 0$ when summing over all types. The Hessian satisfies

$$\boldsymbol{h}^\top \nabla^2 \bar{G}(\boldsymbol{d}) \boldsymbol{h} = \mathbb{E}_{\mu_x}\left[\sum_y \pi_y \left(h_y - \sum_z \pi_z h_z\right)^2\right] > 0.$$

Every π_y is strictly positive, and the extended vector is nonconstant. The weighted variance inside the expectation is therefore positive for every shock realization. Thus $\bar{G}$ has a positive definite Hessian and is strictly convex.

To prove that every interior share vector is attained, fix $\boldsymbol{m}_x \in \Delta^\circ(\mathcal{Y})$ and minimize

$$\bar{G}(\boldsymbol{d}) - \sum_{y \neq y^1} m_{xy} d_y.$$

Set $d_{y^1} = 0$ and $C = \mathbb{E}_{\mu_x} \max_y |\xi_y|$. The preceding lower bound on G yields

$$\begin{aligned}
\bar{G}(\boldsymbol{d}) - \sum_y m_{xy} d_y &\geq \max_y d_y - \sum_y m_{xy} d_y - C \\
&= \sum_y m_{xy} (\max_z d_z - d_y) - C \\
&\geq (\min_y m_{xy})(\max_y d_y - \min_y d_y) - C.
\end{aligned}$$

Since the reference coordinate is zero, $\max_y d_y - \min_y d_y \geq \|\boldsymbol{d}\|_\infty$. Every share is positive, so the objective tends to infinity as $\|\boldsymbol{d}\|_\infty \to \infty$. Continuity gives an attained minimum, and strict convexity makes it unique. Its first-order condition is

$$\nabla \bar{G}(\boldsymbol{d}) = \boldsymbol{m}_{x,-1},$$

where $\boldsymbol{m}_{x,-1}$ contains the nonreference shares. The reference share is also matched because both probability vectors sum to one. Thus the map $\boldsymbol{d} \mapsto \boldsymbol{S}((0,\boldsymbol{d});\boldsymbol{Q}_{x\cdot},\kappa)$ is a bijection onto the interior simplex. Write the uniquely recovered indices as $\boldsymbol{\delta}(\boldsymbol{m}_x;\boldsymbol{Q}_{x\cdot},\kappa)$, with $\delta_{y^1}=0$, and define

$$g_y^*(\boldsymbol{m}_x;\boldsymbol{Q}_{x\cdot},\kappa)=\kappa\delta_y(\boldsymbol{m}_x;\boldsymbol{Q}_{x\cdot},\kappa), \qquad g_{y^1}^*=0.$$

The preceding minimization is the convex-conjugate construction of the inverse, following the approach of Galichon and Salanié (2022).

At the true cost, the indices generating (5.7) are $Q_{xy}U_{xy}/k_x+\log\tilde{p}_{xy}$. Subtracting their reference component leaves the shares unchanged. Uniqueness of the normalized inverse therefore implies

$$\delta_y(\boldsymbol{m}_x;\boldsymbol{Q}_{x\cdot},k_x)=\frac{Q_{xy}U_{xy}-Q_{xy^1}U_{xy^1}}{k_x}+\log\frac{\tilde{p}_{xy}}{\tilde{p}_{xy^1}}.$$

Multiplication by k_x gives (5.8).

Step 2. Differentiability in the candidate cost. Let $B_y=Q_{xy}(\varepsilon_y-\varepsilon_0)$, so $\xi_y(\kappa)=B_y/\kappa$. Holding the indices fixed,

$$\frac{\partial\pi_y}{\partial\kappa}=-\frac{\pi_y}{\kappa^2}\left(B_y-\sum_z\pi_z B_z\right).$$

On any compact interval $\kappa\in[\underline{\kappa},\overline{\kappa}]\subset(0,\infty)$, the absolute value of this derivative is bounded by $2\max_z|B_z|/\underline{\kappa}^2$, whose expectation is finite by Assumption 2.1. Dominated convergence gives a continuously differentiable share map in $(\boldsymbol{d},\kappa)$. Apply the implicit function theorem to the reduced system

$$\nabla_{\boldsymbol{d}}\bar{G}(\boldsymbol{d};\boldsymbol{Q}_{x\cdot},\kappa)-\boldsymbol{m}_{x,-1}=\boldsymbol{0}.$$

Its Jacobian with respect to $\boldsymbol{d}$ is the positive definite Hessian $\nabla^2_{\boldsymbol{d}}\bar{G}$. The theorem therefore yields a continuously differentiable inverse in κ, locally at every positive cost. Uniqueness makes these local inverses agree on $(0,\infty)$. Multiplication by κ shows that g_y^* is also continuously differentiable in the candidate cost.

Step 3. Identification up to the reference payoff. Fix an admissible parameter vector that generates the observed data, and denote its cost by k_x. Hold its reference payoff U_{xy^1} fixed when comparing observationally equivalent vectors. For each positive candidate cost κ, Step 1 uniquely determines the normalized indices that reproduce each environment's shares. Solving (5.8) for utility gives exactly $U_{xy,t}(\kappa)$ in (5.9). Thus a candidate with this reference payoff and common utilities reproduces both sets of shares if and only if

$$H_y(\kappa)=0 \quad \text{for every } y\neq y^1, \qquad U_{xy}=U_{xy,1}(\kappa),$$

and the implied utilities satisfy the maintained admissibility restrictions. In particular, $H_y(k_x) = 0$ for every $y \neq y^1$. This establishes the correspondence between admissible roots and observationally equivalent parameter vectors with the same reference payoff.

Step 2 makes each H_y continuously differentiable. If $H'_y(k_x) \neq 0$ for some y, its derivative keeps the same sign in a sufficiently small neighborhood of k_x. That H_y is strictly monotone in the neighborhood and has its only zero there at k_x. Any common zero of all the H_y must also be a zero of this component. Hence the cost is locally unique for the given reference payoff, and (5.9) uniquely recovers the remaining utilities. This proves local identification up to U_{xy^1}. If the common root is globally unique among admissible candidates, the same reasoning proves global identification up to U_{xy^1}.

Step 4. Equivalence with the rank condition. For completeness, the derivative condition is equivalent to full column rank of the joint restrictions in $(\kappa, \boldsymbol{U}_{-1})$. Write them as

$$R_{yt}(\kappa, \boldsymbol{U}_{-1}) = Q_{xy,t}\big[U_{xy,t}(\kappa) - U_{xy}\big] = 0, \qquad y \neq y^1, \quad t = 1, 2,$$

where $\boldsymbol{U}_{-1} = (U_{xy})_{y \neq y^1}$. A vector $(h_0, (h_y)_{y \neq y^1})$ lies in the Jacobian's null space at the truth if and only if

$$U'_{xy,t}(k_x)h_0 - h_y = 0 \quad \text{for every } y, t,$$

because $Q_{xy,t} > 0$. Subtracting the two equations for each y gives $H'_y(k_x)h_0 = 0$. If any $H'_y(k_x)$ is nonzero, then $h_0 = 0$ and all $h_y = 0$, so the Jacobian has full column rank. If every $H'_y(k_x) = 0$, the nonzero vector with $h_0 = 1$ and $h_y = U'_{xy,1}(k_x)$ belongs to the null space. This proves equivalence with the stated rank condition. □

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